



NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission

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TITLE: HR Employee Leave Management Tool

GITHUB LINK : [https://github.com/ subha271507-tech](https://github.com/subha271507-tech)

1. ABSTRACT

The HR Employee Leave Management Tool is a web-based application developed using React.js to simplify the process of managing employee leave requests within an organization. Employees can submit leave applications through an easy-to-use form, while HR administrators can review, approve, or reject requests. The application maintains leave records, displays leave history, and provides filtering options to view requests by employee. By automating leave tracking and status updates, the system improves transparency, reduces manual paperwork, and enhances workforce management efficiency.

2. PROBLEM STATEMENT

Many organizations still rely on spreadsheets, emails, or manual registers to manage employee leave requests. This approach often leads to data inconsistency, delays in approval, and difficulty tracking leave history. There is a need for a centralized and user-friendly system that enables employees to submit leave requests and allows HR personnel to efficiently manage approvals and records.

3.OBJECTIVES

- ❖ To develop a responsive Single Page Application (SPA) using React.js.
- ❖ To provide employees with an online leave application form.
- ❖ To enable HR administrators to approve or reject leave requests.
- ❖ To display leave request status as Approved, Rejected, or Pending.
- ❖ To implement employee-based filtering for quick record access.
- ❖ To maintain and display complete leave history records.

4.PROJECT SCOPE

- ❖ In-Scope Features
- ❖ Employee leave application form.
- ❖ Leave status tracking.
- ❖ HR approval/rejection management.
- ❖ Employee-wise leave filtering.
- ❖ Leave history display.
- ❖ Responsive user interface.

5. TECHNOLOGIES USED

- ❖ Frontend Framework: React.js
- ❖ State Management: React Hooks (useState, useEffect)
- ❖ Styling: CSS / Inline CSS / Bootstrap
- ❖ Development Environment: Node.js and npm
- ❖ Data Storage: Local State / Local Storage (for demo version)

6. APPLICATION FLOW

- ❖ Employee Side
- ❖ Employee opens the application.
- ❖ Employee fills out the leave application form.
- ❖ Employee submits the request.
- ❖ Leave request is saved with Pending status.
- ❖ HR Side
- ❖ HR views all submitted leave requests.
- ❖ HR filters requests by employee.
- ❖ HR reviews request details.
- ❖ HR approves or rejects the request.
- ❖ Status is updated instantly.

7. KEY REACT CONCEPTS USED

- ❖ useState Hook
- ❖ Used to manage:
 - ❖ Employee leave data.
 - ❖ Leave status updates.
 - ❖ Form inputs.
 - ❖ Filter values.
- ❖ useEffect Hook
- ❖ Used for:
 - ❖ Loading stored leave records.
 - ❖ Updating data when changes occur.
- ❖ Conditional Rendering
- ❖ Used to display:
 - ❖ Approved status.
 - ❖ Rejected status.
 - ❖ Pending status.
- ❖ List Rendering
- ❖ Used to:
 - ❖ Display leave records dynamically.
 - ❖ Render employee leave history using .map().
- ❖ Event Handling
- ❖ Used for:
 - ❖ Form submission.
 - ❖ Approve/Reject actions.
 - ❖ Employee filtering.



8. CHALLENGES FACED

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Challenge 1: Managing Status Updates

Updating leave status without affecting other records.

Solution:

Implemented unique IDs for each leave request and updated records using array mapping techniques.

Challenge 2: Filtering Employee Records

Displaying leave requests for a selected employee only.

Solution:

Used React's filtering methods to dynamically display matching records.

Challenge 3: Form Validation

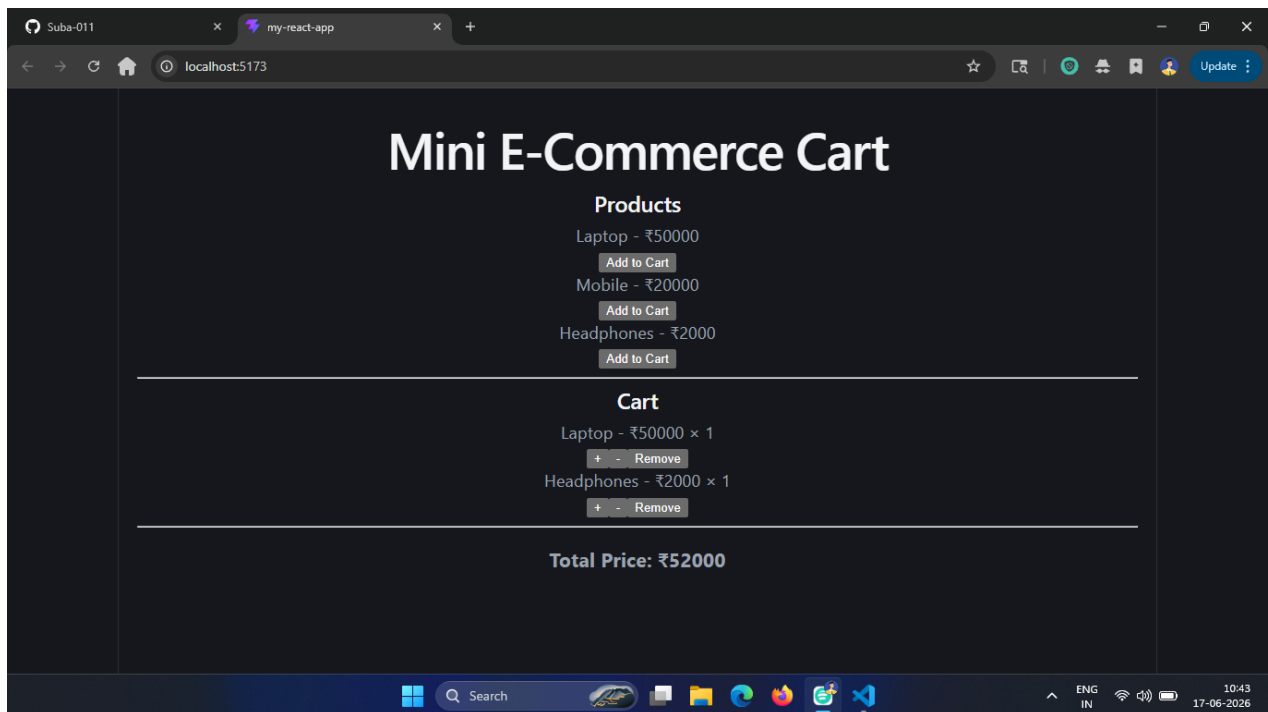
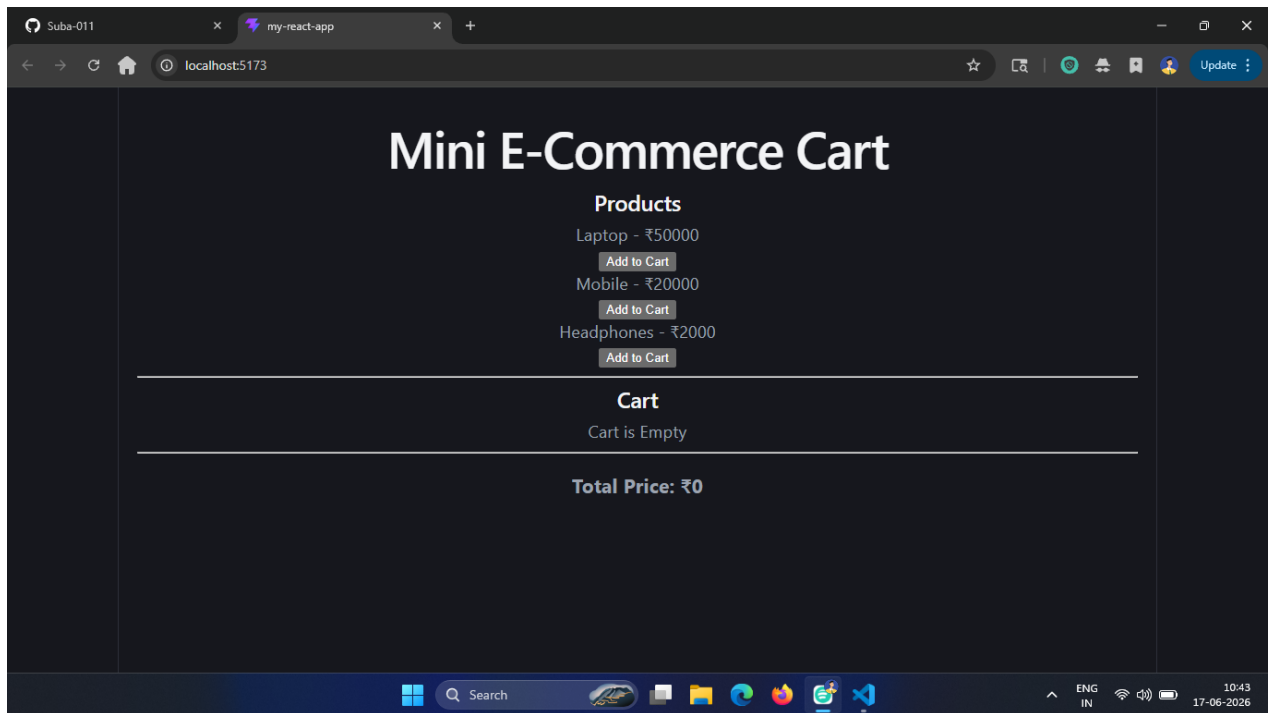
Preventing incomplete leave submissions.

Solution:

Added validation checks before allowing form submission.

project delivers an efficient solution for HR operations while enhancing user experience and organizational productivity.

9. EXPECTED OUTPUT



10. LEARNING OUTCOMES

- ❖ Learned React component-based architecture.
- ❖ Gained experience with React Hooks.
- ❖ Implemented CRUD-like operations using state management.
- ❖ Developed skills in form handling and validation.
- ❖ Learned dynamic filtering and conditional rendering techniques.
- ❖ Improved UI development and data management skills.

11. CONCLUSION

The HR Employee Leave Management Tool successfully streamlines the leave request and approval process within an organization. The application demonstrates the practical use of React.js in building responsive and interactive business management systems. By providing leave application, approval workflow, employee filtering, and leave history tracking, the project delivers an efficient solution for HR operations while enhancing user experience and organizational productivity.